

Submit 8 problems total from the union of *LADW* Ch 3.1-6 (copied below) and the supplementary problems. Please label all *LADW* problems by “Chapter.Section.Exercise,” e.g. 1.3.4. Label the supplementary problems as they’re labeled on the handout. Please make a note about any collaboration or use of outside resources. Please put a “G” in front of the four problems you would like graded for correctness. Additionally, put an “F” in front of any of the problems graded for completion on which you would like written feedback from the grader.

Suggested *LADW* problems: 3.3.5, 3.3.9, 3.3.10, 3.4.1, 3.4.2, 3.5.5, 3.7.3

Section 3.1 *No exercises in this section.*

Section 3.2: *No exercises in this section.*

Section 3.3

Exercises.

X 3.1. If A is an $n \times n$ matrix, how are the determinants $\det A$ and $\det(5A)$ related?
Remark: $\det(5A) = 5 \det A$ only in the trivial case of 1×1 matrices

3.2. How are the determinants $\det A$ and $\det B$ related if

$$\text{a) } A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \quad B = \begin{pmatrix} 2a_1 & 3a_2 & 5a_3 \\ 2b_1 & 3b_2 & 5b_3 \\ 2c_1 & 3c_2 & 5c_3 \end{pmatrix};$$

$$\text{b) } A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \quad B = \begin{pmatrix} 3a_1 & 4a_2 + 5a_1 & 5a_3 \\ 3b_1 & 4b_2 + 5b_1 & 5b_3 \\ 3c_1 & 4c_2 + 5c_1 & 5c_3 \end{pmatrix}.$$

X 3.3. Using column or row operations compute the determinants

$$\begin{vmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{vmatrix}, \quad \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}, \quad \begin{vmatrix} 1 & 0 & -2 & 3 \\ -3 & 1 & 1 & 2 \\ 0 & 4 & -1 & 1 \\ 2 & 3 & 0 & 1 \end{vmatrix}, \quad \begin{vmatrix} 1 & x \\ 1 & y \end{vmatrix}.$$

3.4. A square ($n \times n$) matrix is called skew-symmetric (or antisymmetric) if $A^T = -A$. Prove that if A is skew-symmetric and n is odd, then $\det A = 0$. Is this true for even n ?

3.5. A square matrix is called *nilpotent* if $A^k = \mathbf{0}$ for some positive integer k . Show that for a nilpotent matrix A $\det A = 0$.

3.6. Prove that if the matrices A and B are similar, then $\det A = \det B$.

3.7. A real square matrix Q is called orthogonal if $Q^T Q = I$. Prove that if Q is an orthogonal matrix then $\det Q = \pm 1$.

3.8. Show that

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = (z-x)(z-y)(y-x).$$

This is a particular case of the so-called Vandermonde determinant.

3.9. Let points A, B and C in the plane $\mathbb{R}^2$ have coordinates (x_1, y_1) , (x_2, y_2) and (x_3, y_3) respectively. Show that the area of triangle ABC is the absolute value of

$$\frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}.$$

Hint: use row operations and geometric interpretation of 2×2 determinants (area).

3.10. Let A be a square matrix. Show that block triangular matrices

$$\begin{pmatrix} I & * \\ \mathbf{0} & A \end{pmatrix}, \quad \begin{pmatrix} A & * \\ \mathbf{0} & I \end{pmatrix}, \quad \begin{pmatrix} I & \mathbf{0} \\ * & A \end{pmatrix}, \quad \begin{pmatrix} A & \mathbf{0} \\ * & I \end{pmatrix}$$

all have determinant equal to $\det A$. Here $*$ can be anything.

The following problems illustrate the power of block matrix notation.

3.11. Use the previous problem to show that if A and C are square matrices, then

$$\det \begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \det A \det C.$$

Hint: $\begin{pmatrix} A & B \\ \mathbf{0} & C \end{pmatrix} = \begin{pmatrix} I & B \\ \mathbf{0} & C \end{pmatrix} \begin{pmatrix} A & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix}.$

3.12. Let A be $m \times n$ and B be $n \times m$ matrices. Prove that

$$\det \begin{pmatrix} 0 & A \\ -B & I \end{pmatrix} = \det(AB).$$

Hint: While it is possible to transform the matrix by row operations to a form where the determinant is easy to compute, the easiest way is to right multiply the matrix by $\begin{pmatrix} I & 0 \\ B & I \end{pmatrix}.$

Section 3.4

Exercises.

4.1. Suppose the permutation σ takes $(1, 2, 3, 4, 5)$ to $(5, 4, 1, 2, 3)$.

- Find sign of σ ;
- What does $\sigma^2 := \sigma \circ \sigma$ do to $(1, 2, 3, 4, 5)$?
- What does the inverse permutation σ^{-1} do to $(1, 2, 3, 4, 5)$?
- What is the sign of σ^{-1} ?

4.2. Let P be a *permutation matrix*, i.e. an $n \times n$ matrix consisting of zeroes and ones and such that there is exactly one 1 in every row and every column.

- Can you describe the corresponding linear transformation? That will explain the name.
- Show that P is invertible. Can you describe P^{-1} ?
- Show that for some $N > 0$

$$P^N := \underbrace{PP \dots P}_{N \text{ times}} = I.$$

Use the fact that there are only finitely many permutations.

4.3. Why is there an even number of permutations of $(1, 2, \dots, 9)$ and why are exactly half of them odd permutations? **Hint:** This problem can be hard to solve in terms of permutations, but there is a very simple solution using determinants.

4.4. If σ is an odd permutation, explain why σ^2 is even but σ^{-1} is odd.

4.5. How many multiplications and additions is required to compute the determinant using formal definition (4.2) of the determinant of an $n \times n$ matrix? Do not count the operations needed to compute sign σ .

Section 3.5

Exercises.

X 5.1. Evaluate the determinants using any method

$$\begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & -5 \\ 6 & -4 & 3 \end{vmatrix}, \quad \begin{vmatrix} 1 & -2 & 3 & -12 \\ -5 & 12 & -14 & 19 \\ -9 & 22 & -20 & 31 \\ -4 & 9 & -14 & 15 \end{vmatrix}.$$

X 5.2. Use row (column) expansion to evaluate the determinants. Note, that you don't need to use the first row (column): picking row (column) with many zeroes will simplify your calculations.

$$\begin{vmatrix} 1 & 2 & 0 \\ 1 & 1 & 5 \\ 1 & -3 & 0 \end{vmatrix}, \quad \begin{vmatrix} 4 & -6 & -4 & 4 \\ 2 & 1 & 0 & 0 \\ 0 & -3 & 1 & 3 \\ -2 & 2 & -3 & -5 \end{vmatrix}.$$

5.3. For the $n \times n$ matrix

$$A = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 & a_0 \\ -1 & 0 & 0 & \dots & 0 & a_1 \\ 0 & -1 & 0 & \dots & 0 & a_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & a_{n-2} \\ 0 & 0 & 0 & \dots & -1 & a_{n-1} \end{pmatrix}$$

compute $\det(A + tI)$, where I is $n \times n$ identity matrix. You should get a nice expression involving $a_0, a_1, \dots, a_{n-1}$ and t . Row expansion and induction is probably the best way to go.

X 5.4. Using cofactor formula compute inverses of the matrices

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} 19 & -17 \\ 3 & -2 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 3 & 5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 & 0 \\ 2 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$

5.5. Let D_n be the determinant of the $n \times n$ tridiagonal matrix

$$\begin{pmatrix} 1 & -1 & & & 0 \\ 1 & 1 & -1 & & \\ & 1 & \ddots & \ddots & \\ & & \ddots & 1 & -1 \\ 0 & & & 1 & 1 \end{pmatrix}.$$

Using cofactor expansion show that $D_n = D_{n-1} + D_{n-2}$. This yields that the sequence D_n is the Fibonacci sequence 1, 2, 3, 5, 8, 13, 21, ...

5.6. Vandermonde determinant revisited. Our goal is to prove the formula

$$\begin{vmatrix} 1 & c_0 & c_0^2 & \dots & c_0^n \\ 1 & c_1 & c_1^2 & \dots & c_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & c_n & c_n^2 & \dots & c_n^n \end{vmatrix} = \prod_{0 \leq j < k \leq n} (c_k - c_j)$$

for the $(n+1) \times (n+1)$ Vandermonde determinant.

We will apply induction. To do this

- Check that the formula holds for $n = 1, n = 2$.
- Call the variable c_n in the last row x , and show that the determinant is a polynomial of degree n , $A_0 + A_1x + A_2x^2 + \dots + A_nx^n$, with the coefficients A_k depending on $c_0, c_1, \dots, c_{n-1}$.
- Show that the polynomial has zeroes at $x = c_0, c_1, \dots, c_{n-1}$, so it can be represented as $A_n \cdot (x - c_0)(x - c_1) \dots (x - c_{n-1})$, where A_n as above.
- Assuming that the formula for the Vandermonde determinant is true for $n - 1$, compute A_n and prove the formula for n .

5.7. How many multiplication is needed to compute the determinant of an $n \times n$ matrix using the cofactor expansion? Prove the formula.

Section 3.6: No exercises in this section.

Section 3.7

7.1. True or false

- Determinant is only defined for square matrices.
- If two rows or columns of A are identical, then $\det A = 0$.
- If B is the matrix obtained from A by interchanging two rows (or columns), then $\det B = \det A$.
- If B is the matrix obtained from A by multiplying a row (column) of A by a scalar α , then $\det B = \det A$.
- If B is the matrix obtained from A by adding a multiple of a row to some other row, then $\det B = \det A$.
- The determinant of a triangular matrix is the product of its diagonal entries.
- $\det(A^T) = -\det(A)$.
- $\det(AB) = \det(A)\det(B)$.
 - A matrix A is invertible if and only if $\det A \neq 0$.
 - If A is an invertible matrix, then $\det(A^{-1}) = 1/\det(A)$.

7.2. Let A be an $n \times n$ matrix. How are $\det(3A)$, $\det(-A)$ and $\det(A^2)$ related to $\det A$.

7.3. If the entries of both A and A^{-1} are integers, is it possible that $\det A = 3$?
Hint: what is $\det(A)\det(A^{-1})$?

7.4. Let $\mathbf{v}_1, \mathbf{v}_2$ be vectors in $\mathbb{R}^2$ and let A be the 2×2 matrix with columns $\mathbf{v}_1, \mathbf{v}_2$. Prove that $|\det A|$ is the area of the parallelogram with two sides given by the vectors $\mathbf{v}_1, \mathbf{v}_2$.

Consider first the case when $\mathbf{v}_1 = (x_1, 0)^T$. To treat general case $\mathbf{v}_1 = (x_1, y_1)^T$ left multiply A by a rotation matrix that transforms vector $\mathbf{v}_1$ into $(\tilde{x}_1, 0)^T$. **Hint:** what is the determinant of a rotation matrix?

The following problem illustrates relation between the sign of the determinant and the so-called *orientation* of a system of vectors.

7.5. Let $\mathbf{v}_1, \mathbf{v}_2$ be vectors in $\mathbb{R}^2$. Show that $D(\mathbf{v}_1, \mathbf{v}_2) > 0$ if and only if there exists a rotation T_α such that the vector $T_\alpha \mathbf{v}_1$ is parallel to $\mathbf{e}_1$ (and looking in the same direction), and $T_\alpha \mathbf{v}_2$ is in the upper half-plane $x_2 > 0$ (the same half-plane as $\mathbf{e}_2$).

Hint: What is the determinant of a rotation matrix?

Supplementary problems for HW4

I. Consider the matrices in $M_{2 \times 2}^{\mathbb{Z}/5\mathbb{Z}}$. What are the possible values for their determinants? How many of the matrices have determinant equal to each of those values? Explain your thinking.

II. Always sometimes, or never? (If sometimes, give an example each way. If always or never, explain why.)

- A real square matrix with positive entries has a positive determinant.
- A real square matrix whose determinant is an integer also has all integer entries.
- A real square matrix with all integer entries has an integer for its determinant.

III. Let $f(x) = \det A$ where

$$A = \begin{pmatrix} -6 & 9 & -2 & 1 & 3 \\ 4 & 0 & 7 & 1 & -5 \\ 4 & 0 & 0 & -8 & 8 \\ x & 9 & -3 & -8 & -9 \\ 4 & 0 & 0 & 0 & -2 \end{pmatrix}.$$

Find $f'(x)$.

IV. Let C be the 100×100 matrix with each entry equalling 1, let I be the 100×100 identity matrix, and let D be the diagonal matrix whose $a_{i,i}$ entry is the integer i , where i ranges from 1 to 100. Compute the following: $\det(C)$, $\det D$, $\det(C - I)$, $\det(C - I + D)$. Justify your answers.

V. Consider the set $M_{0,1}(n)$ of $n \times n$ real matrices that have 0 or 1 for each of their entries. The set $M_{0,1}(n)$ contains exactly $2^{(n^2)}$ matrices. Feel free to use software as you explore the following questions.

- What is the biggest determinant that a matrix in $M_{0,1}(2)$ can have? How many of these maximizing matrices are there?
- Same questions, but for $M_{0,1}(3)$?
- What can you say about the maximal determinant and the maximal matrices in $M_{0,1}(n)$ for arbitrary n ? (Note: Anything you can say is great, since this is an unsolved problem in general!)